

Análisis de expresión diferencial - clase

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Librerías

```
# install.packages("BiocManager")

# install.packages("tidyverse")
# install.packages("ggplot2")
# BiocManager::install("DESeq2")
# BiocManager::install("clusterProfiler")

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(DESeq2)
```

```
## Loading required package: S4Vectors
## Loading required package: stats4
## Loading required package: BiocGenerics
##
## Attaching package: 'BiocGenerics'
##
## The following objects are masked from 'package:lubridate':
##
##   intersect, setdiff, union
##
## The following objects are masked from 'package:dplyr':
##
##   combine, intersect, setdiff, union
##
```

```

## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs
##
## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##   match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##   Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##   table, tapply, union, unique, unsplit, which.max, which.min
##
##
## Attaching package: 'S4Vectors'
##
## The following objects are masked from 'package:lubridate':
##
##   second, second<-
##
## The following objects are masked from 'package:dplyr':
##
##   first, rename
##
## The following object is masked from 'package:tidyr':
##
##   expand
##
## The following object is masked from 'package:utils':
##
##   findMatches
##
## The following objects are masked from 'package:base':
##
##   expand.grid, I, unname
##
## Loading required package: IRanges
##
## Attaching package: 'IRanges'
##
## The following object is masked from 'package:lubridate':
##
##   %within%
##
## The following objects are masked from 'package:dplyr':
##
##   collapse, desc, slice
##
## The following object is masked from 'package:purrr':
##
##   reduce
##
## Loading required package: GenomicRanges
## Loading required package: GenomeInfoDb

```

```

## Loading required package: SummarizedExperiment
## Loading required package: MatrixGenerics
## Loading required package: matrixStats
##
## Attaching package: 'matrixStats'
##
## The following object is masked from 'package:dplyr':
##
##     count
##
## Attaching package: 'MatrixGenerics'
##
## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars
##
## Loading required package: Biobase
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname)".
##
## Attaching package: 'Biobase'
##
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
##
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians

```

```
library(clusterProfiler)
```

```

##
## clusterProfiler v4.14.6 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##

```

```
## Please cite:
##
## S Xu, E Hu, Y Cai, Z Xie, X Luo, L Zhan, W Tang, Q Wang, B Liu, R Wang,
## W Xie, T Wu, L Xie, G Yu. Using clusterProfiler to characterize
## multiomics data. Nature Protocols. 2024, 19(11):3292-3320
##
## Attaching package: 'clusterProfiler'
##
## The following object is masked from 'package:IRanges':
##
##     slice
##
## The following object is masked from 'package:S4Vectors':
##
##     rename
##
## The following object is masked from 'package:purrr':
##
##     simplify
##
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(msigdb)
```

Cargar datos

```
rna <- read.table(file="../input/coad_rna.tsv", sep="\t", header=TRUE, check.names = FALSE, row.names =
metadata <- read.table(file="../input/coad_metadata.tsv", sep="\t", header = TRUE)

head(metadata)
```

```
##      PATIENT_ID      SAMPLE_ID SUBTYPE AGE  SEX  OS_STATUS
## 1 TCGA-A6-5667 TCGA-A6-5667-01    CIN  40 Female  0:LIVING
## 2 TCGA-AY-4070 TCGA-AY-4070-01    CIN  50 Female  1:DECEASED
## 3 TCGA-CM-6168 TCGA-CM-6168-01     GS  84 Female  0:LIVING
## 4 TCGA-CA-6716 TCGA-CA-6716-01    CIN  65   Male  0:LIVING
## 5 TCGA-F4-6569 TCGA-F4-6569-01    CIN  60   Male  0:LIVING
## 6 TCGA-A6-A56B TCGA-A6-A56B-01    CIN  57   Male  1:DECEASED
##  COUNTRY_OF_RESIDENCE  SITE_OF_BIOPSY purity ploidy wholeGenomeDuplication
## 1                <NA>      Sigmoid colon  0.69  3.20                true
## 2                <NA>       Cecum      0.31  4.39                true
## 3      United States  Ascending colon  0.35  2.17                false
## 4      Vietnam      Ascending colon  0.73  3.74                true
## 5      Russia      Transverse colon  0.40  2.16                false
## 6      United States  Sigmoid colon  0.72  3.21                true
##      CMS metastasis
## 1      CMS2      false
## 2 no_label      false
## 3      CMS4      false
```

```
## 4      CMS2      false
## 5      CMS4      false
## 6 no_label      false
```

```
dim(rna)
```

```
## [1] 60660   300
```

Extraer el dato a comparar

```
# El modelo DESeq2 necesita una colData que tenga en las filas las muestras, y la columna con el valor
```

```
coldata <- metadata %>% select(PATIENT_ID, wholeGenomeDuplication) %>%
  column_to_rownames(var = "PATIENT_ID") %>%
  mutate(wholeGenomeDuplication = as.factor(wholeGenomeDuplication))
```

```
table(coldata)
```

```
## wholeGenomeDuplication
## false  true
##   178   122
```

```
# eliminar genes sin expresion
```

```
table(rowSums(rna) > 0) # TRUE genes con expresion, FALSE genes donde en todas las muestras son 0
```

```
##
## FALSE  TRUE
## 12252 48408
```

```
rna <- rna[rowSums(rna) > 0, ]
dim(rna)
```

```
## [1] 48408   300
```

Correr análisis de expresión

```
dds <- DESeqDataSetFromMatrix(countData = rna,
                              colData = coldata,
                              design = ~wholeGenomeDuplication)
```

```
dds <- DESeq(dds)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates

## mean-dispersion relationship

## final dispersion estimates

## fitting model and testing

## -- replacing outliers and refitting for 1599 genes
## -- DESeq argument 'minReplicatesForReplace' = 7
## -- original counts are preserved in counts(dds)

## estimating dispersions

## fitting model and testing
```

```
resultsNames(dds)
```

```
## [1] "Intercept"
## [2] "wholeGenomeDuplication_true_vs_false"
```

```
res <- results(dds, name = "wholeGenomeDuplication_true_vs_false") %>% as.data.frame()
summary(res)
```

```
##      baseMean      log2FoldChange      lfcSE      stat
## Min.      : 0.00      Min.      :-4.88285      Min.      :0.0000      Min.      : -8.76153
## 1st Qu.: 0.02      1st Qu.: -0.11598      1st Qu.: 0.1368      1st Qu.: -0.27856
## Median : 0.21      Median : -0.02934      Median : 0.4921      Median : -0.01438
## Mean    : 20.69      Mean    : -0.03760      Mean    : 1.2213      Mean    : -0.05983
## 3rd Qu.: 4.30      3rd Qu.: 0.04479      3rd Qu.: 2.9468      3rd Qu.: 0.09226
## Max.    :33423.21      Max.    : 5.25742      Max.    : 2.9468      Max.    :10.86909
##
##      pvalue      padj
## Min.      :0.0000      Min.      :0.000
## 1st Qu.:0.2494      1st Qu.:0.132
## Median :0.8512      Median :0.454
## Mean    :0.6435      Mean    :0.458
## 3rd Qu.:0.9841      3rd Qu.:0.765
## Max.    :1.0000      Max.    :1.000
##
##      NA's      :25371
```

```
# log2FC > 0 - enriquecido en true, < 0 enriquecido en false
# Un log2FC de 1 indica un cambio de dos veces. Es decir, si es 1 entonces en esa condicion esta expres
true_enriched <- res %>% filter(log2FoldChange > 1, padj < 0.1)
false_enriched <- res %>% filter(log2FoldChange < -1, padj < 0.1)

cat("Hay", nrow(true_enriched), "genes sobreexpresados en duplicacion de genoma completo, versus", nrow
```

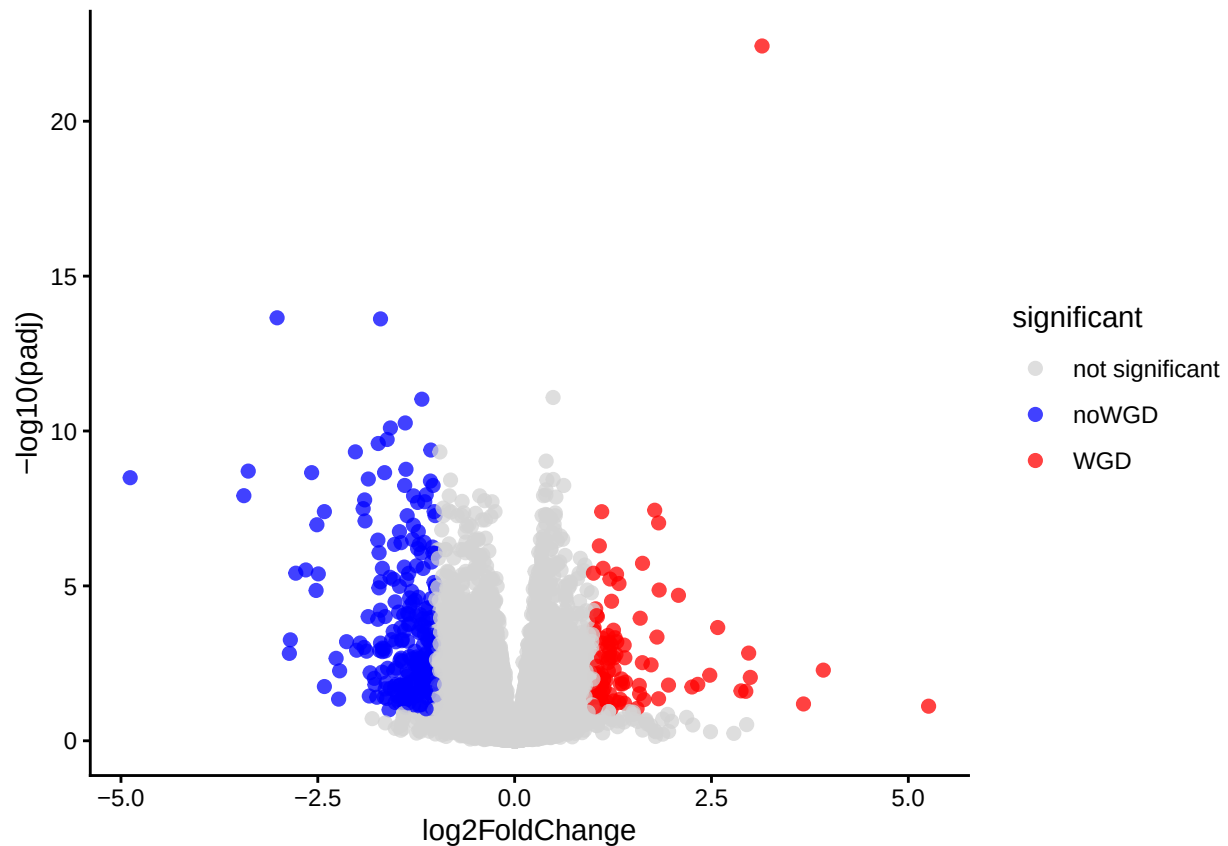
```
## Hay 101 genes sobreexpresados en duplicacion de genoma completo, versus 246 subexpresados.
```

Volcanoplot

```
# Definir colores
res <- res %>%
  mutate(gene = rownames(res),
         significant = ifelse(gene %in% rownames(true_enriched), "WGD",
                              ifelse(gene %in% rownames(false_enriched),
                                     "noWGD",
                                     "not significant"))))

# generar volcano
ggplot(res, aes(x = log2FoldChange, y = -log10(padj),
               color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("WGD" = "red",
                               "noWGD" = "blue",
                               "not significant" = "lightgray")) +
  theme_classic()
```

```
## Warning: Removed 25371 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```

# Definir una base de datos. Por ejemplo = GO procesos biológicos
go_bp <- msigdbbr(species = "Homo sapiens",
                  collection = "C5",
                  subcollection = "GO:BP")

# El universo de genes a estudiar debe ser los genes utilizados en la prueba
universo <- res$gene

# Se establece una tabla de asociacion proceso-gen
term2gene <- unique(as.data.frame(go_bp[, c("gs_name", "ensembl_gene"])))

```

```

# lista de genes con grupo
genes <- list("WGD" = res$gene[res$significant == "WGD"],
             "noWGD" = res$gene[res$significant == "noWGD"])

enrich <- compareCluster(geneClusters = genes,
                        fun = "enricher",
                        TERM2GENE = term2gene,
                        universe = universo,
                        pAdjustMethod = "BH",
                        pvalueCutoff = 0.1,
                        minGSSize = 3)

```

```

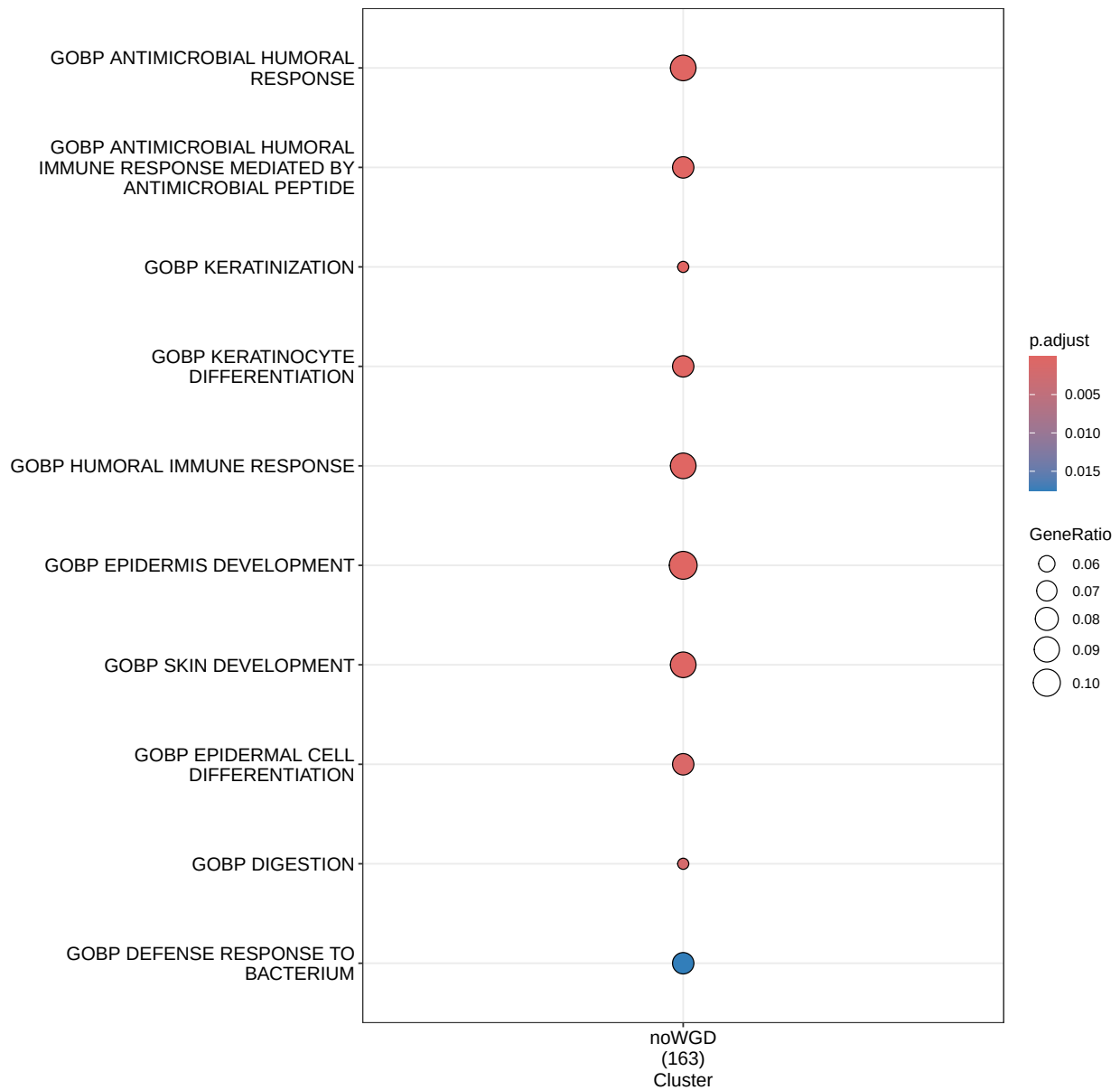
dotplot(enrich, showCategory = 10,
        color = "p.adjust")

```

```

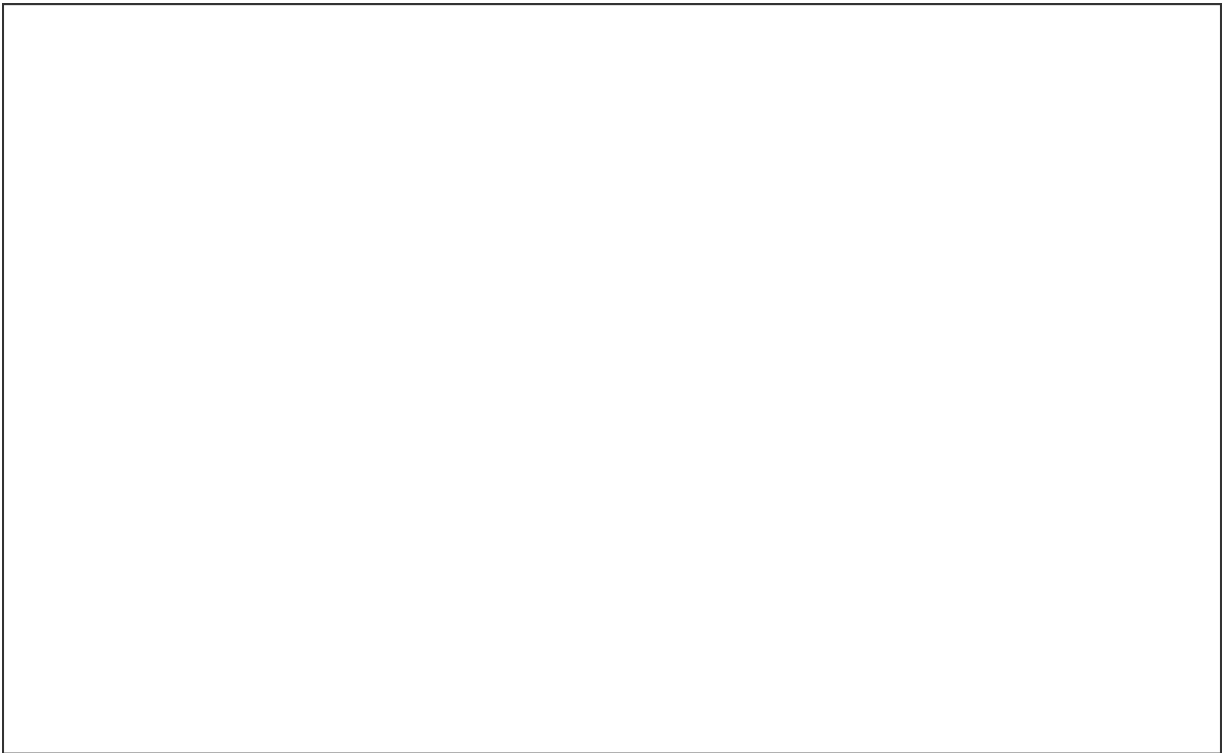
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## i The deprecated feature was likely used in the enrichplot package.
##   Please report the issue at
##   <https://github.com/GuangchuangYu/enrichplot/issues>.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



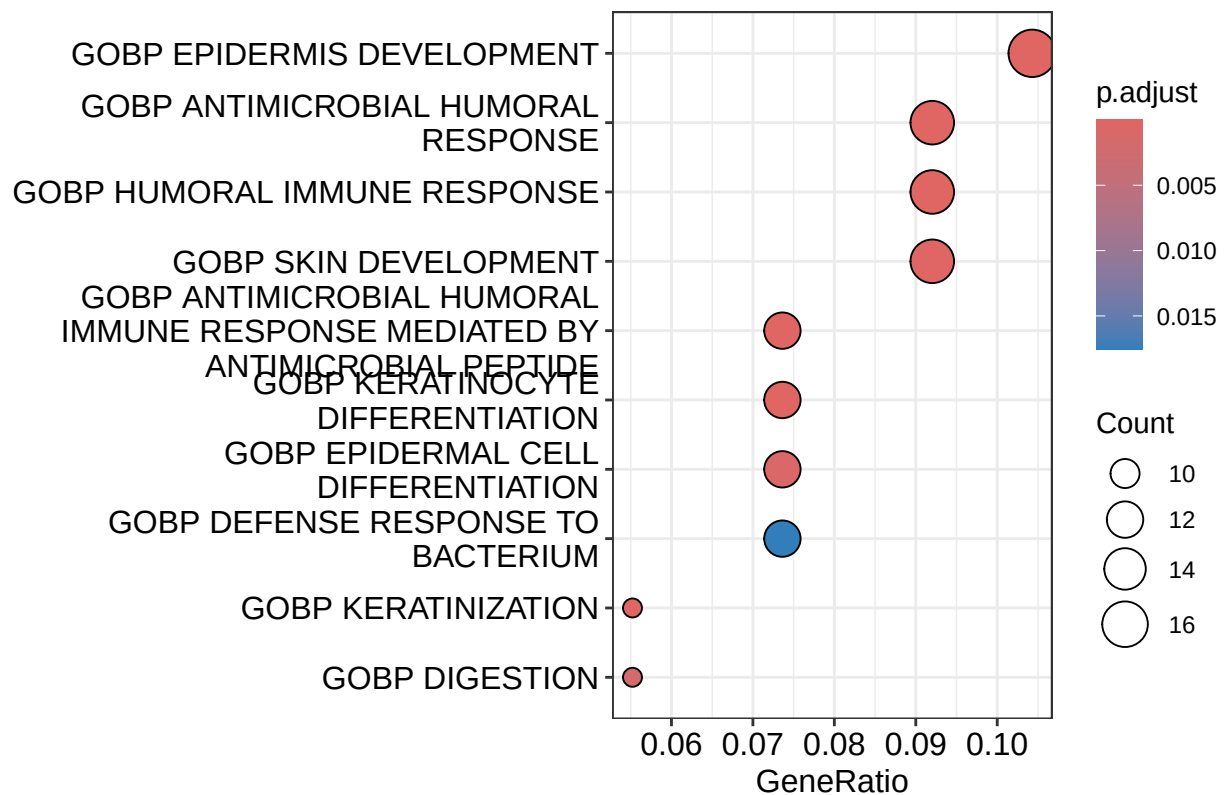
Alternativa: un enriquecimiento por categoria

```
wgd_enrich <- enricher(gene = genes$WGD, TERM2GENE = term2gene,
                        universe = universo, pAdjustMethod = "BH",
                        pvalueCutoff = 0.1, minGSSize = 3)
dotplot(wgd_enrich, showCategory = 10,
        color = "p.adjust")
```



GeneRatio

```
nowgd_enrich <- enricher(gene = genes$noWGD, TERM2GENE = term2gene,  
                          universe = universo, pAdjustMethod = "BH",  
                          pvalueCutoff = 0.1, minGSSize = 3)  
dotplot(nowgd_enrich, showCategory = 10,  
        color = "p.adjust")
```



Parte 2: pureza como covariable

```
coldata <- metadata %>% select(PATIENT_ID, wholeGenomeDuplication, purity) %>%
  column_to_rownames(var = "PATIENT_ID") %>%
  mutate(wholeGenomeDuplication = as.factor(wholeGenomeDuplication))

dds <- DESeqDataSetFromMatrix(countData = rna,
                              colData = coldata,
                              design = ~wholeGenomeDuplication + purity)

dds <- DESeq(dds)

## estimating size factors

## estimating dispersions

## gene-wise dispersion estimates

## mean-dispersion relationship

## final dispersion estimates
```

```
## fitting model and testing

## -- replacing outliers and refitting for 742 genes
## -- DESeq argument 'minReplicatesForReplace' = 7
## -- original counts are preserved in counts(dds)
```

```
## estimating dispersions
```

```
## fitting model and testing
```

```
resultsNames(dds)
```

```
## [1] "Intercept"
## [2] "wholeGenomeDuplication_true_vs_false"
## [3] "purity"
```

```
res_2 <- results(dds, name = "wholeGenomeDuplication_true_vs_false") %>% as.data.frame()
summary(res_2)
```

```
##      baseMean      log2FoldChange      lfcSE      stat
## Min.   : 0.00   Min.   :-20.23000   Min.   :0.02753   Min.   : -11.79306
## 1st Qu.: 0.02   1st Qu.: -0.12784   1st Qu.:0.13605   1st Qu.: -0.29111
## Median : 0.22   Median : -0.03413   Median :0.51596   Median : -0.01444
## Mean   : 20.79   Mean   : -0.03116   Mean   :1.25680   Mean   : -0.03771
## 3rd Qu.: 4.43   3rd Qu.: 0.05079   3rd Qu.:2.99636   3rd Qu.: 0.11548
## Max.   :33423.21   Max.   : 15.84271   Max.   :2.99636   Max.   : 11.52531
##
##      pvalue      padj
## Min.   :0.0000   Min.   :0.000
## 1st Qu.:0.2042   1st Qu.:0.070
## Median :0.8331   Median :0.403
## Mean   :0.6271   Mean   :0.424
## 3rd Qu.:0.9834   3rd Qu.:0.742
## Max.   :1.0000   Max.   :1.000
## NA's   :311     NA's   :24529
```

```
true_enriched <- res_2 %>% filter(log2FoldChange > 1, padj < 0.1)
false_enriched <- res_2 %>% filter(log2FoldChange < -1, padj < 0.1)
```

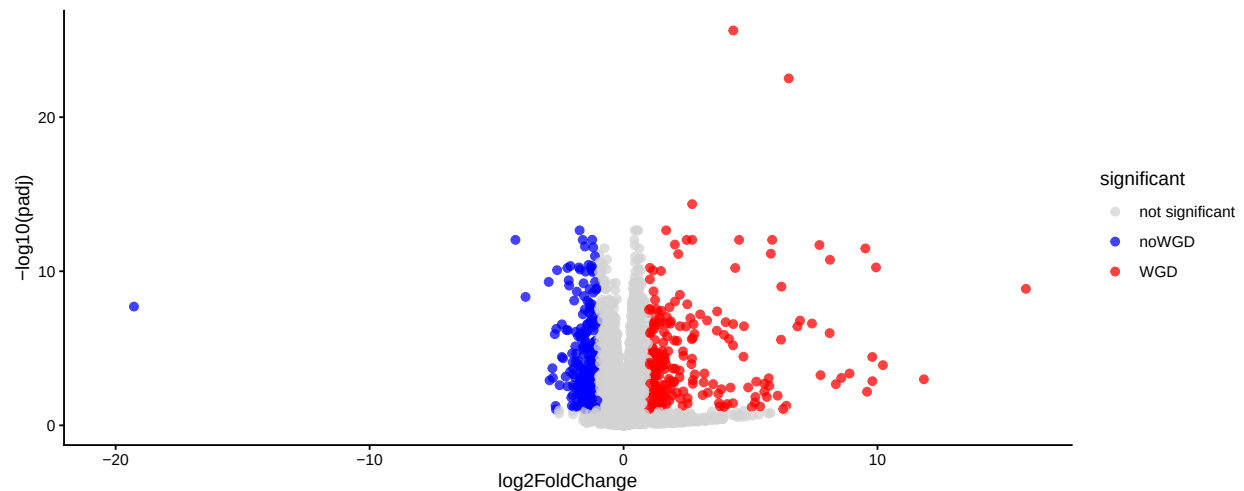
```
cat("Hay", nrow(true_enriched), "genes sobreexpresados en duplicacion de genoma completo, versus", nrow
```

```
## Hay 283 genes sobreexpresados en duplicacion de genoma completo, versus 291 subexpresados.
```

```
res_2 <- res_2 %>%
  mutate(gene = rownames(res_2),
         significant = ifelse(gene %in% rownames(true_enriched), "WGD",
                              ifelse(gene %in% rownames(false_enriched),
                                       "noWGD",
                                       "not significant"))))
```

```
# generar volcano
ggplot(res_2, aes(x = log2FoldChange, y = -log10(padj),
                  color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("WGD" = "red",
                                "noWGD" = "blue",
                                "not significant" = "lightgray")) +
  theme_classic()
```

```
## Warning: Removed 24529 rows containing missing values or values outside the scale range
## ('geom_point()').
```

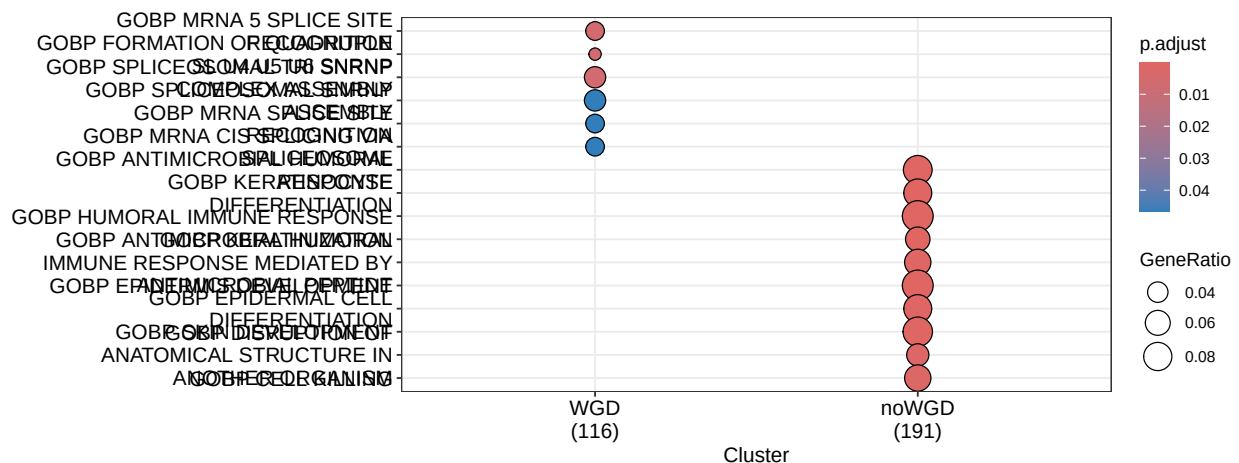


```
universo <- res_2$gene

genes <- list("WGD" = res_2$gene[res_2$significant == "WGD"],
              "noWGD" = res_2$gene[res_2$significant == "noWGD"])

enrich_2 <- compareCluster(geneClusters = genes,
                           fun = "enricher",
                           TERM2GENE = term2gene,
                           universe = universo,
                           pAdjustMethod = "BH",
                           pvalueCutoff = 0.1,
                           minGSSize = 3)

dotplot(enrich_2, showCategory = 10,
        color = "p.adjust")
```



```
res_purity <- results(dds, name = "purity") %>% as.data.frame()
res_purity <- res_purity %>%
  mutate(gene = rownames(res_purity),
         significant = ifelse((log2FoldChange > 1 & padj < 0.1), "purity_high",
                             ifelse((log2FoldChange < -1 & padj < 0.1), "purity_low", "not significant")),
         table(res_purity$significant))
```

```
##
## not significant      purity_high      purity_low
##          40827          2535          4051
```

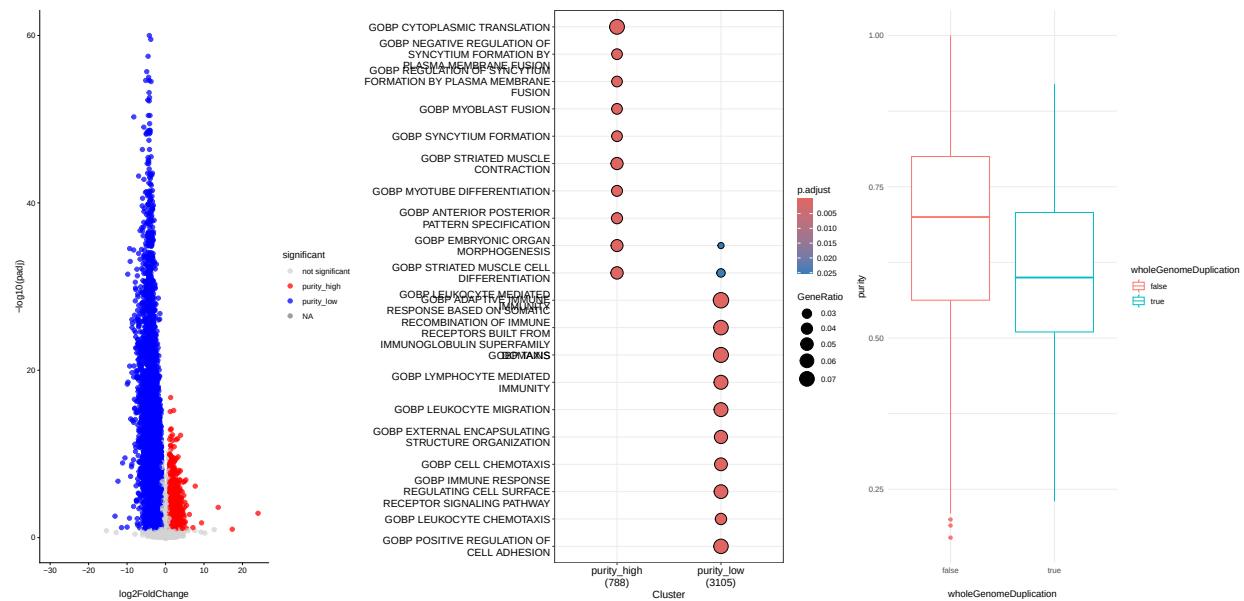
```
universo <- res_purity$gene
genes <- list("purity_high" = res_purity$gene[res_purity$significant == "purity_high"],
             "purity_low" = res_purity$gene[res_purity$significant == "purity_low"])

enrich_purity <- compareCluster(geneClusters = genes,
                               fun = "enricher",
                               TERM2GENE = term2gene,
                               universe = universo,
                               pAdjustMethod = "BH",
                               pvalueCutoff = 0.1,
                               minGSSize = 3)

ggplot(res_purity, aes(x = log2FoldChange, y = -log10(padj),
                      color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("purity_high" = "red",
                                "purity_low" = "blue",
                                "not significant" = "lightgray")) +
  theme_classic() +
dotplot(enrich_purity, showCategory = 10,
        color = "p.adjust") +
ggplot(coldata, aes(x=wholeGenomeDuplication, y = purity,
                   color = wholeGenomeDuplication)) +
  geom_boxplot() + theme_minimal()
```

```
## Warning: Removed 23621 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



Repetir análisis: sexo

```
coldata <- metadata %>% dplyr::select(PATIENT_ID, SEX, purity, AGE) %>%
  column_to_rownames(var = "PATIENT_ID") %>%
  mutate(SEX = as.factor(SEX))

# eliminar muestras con SEX = NA
coldata <- coldata %>%
  filter(!is.na(SEX))

# dejar en rna solo las muestras restantes
rna <- rna[, rownames(coldata)]

dds_sex <- DESeqDataSetFromMatrix(countData = rna,
  colData = coldata,
  design = ~SEX + purity + AGE)
```

```
## the design formula contains one or more numeric variables with integer values,
## specifying a model with increasing fold change for higher values.
## did you mean for this to be a factor? if so, first convert
## this variable to a factor using the factor() function

## the design formula contains one or more numeric variables that have mean or
## standard deviation larger than 5 (an arbitrary threshold to trigger this message).
## Including numeric variables with large mean can induce collinearity with the intercept.
## Users should center and scale numeric variables in the design to improve GLM convergence.
```

```
dds_sex <- DESeq(dds_sex)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
resultsNames(dds_sex)
```

```
## [1] "Intercept"          "SEX_Male_vs_Female" "purity"  
## [4] "AGE"
```

```
res_sex <- results(dds_sex, name = "SEX_Male_vs_Female") %>% as.data.frame()  
summary(res_sex)
```

```
##      baseMean      log2FoldChange      lfcSE      stat  
## Min.      : 0.00      Min.      :-5.519039      Min.      :0.02828      Min.      : -21.936663  
## 1st Qu.: 0.02      1st Qu.: -0.050733      1st Qu.: 0.13435      1st Qu.: -0.124906  
## Median : 0.22      Median : 0.005551      Median : 0.51742      Median : 0.002002  
## Mean   : 20.80      Mean   : 0.026240      Mean   : 1.25022      Mean   : 0.005672  
## 3rd Qu.: 4.43      3rd Qu.: 0.072818      3rd Qu.: 2.94603      3rd Qu.: 0.153588  
## Max.   :33367.81      Max.   : 7.241181      Max.   : 2.94603      Max.   : 28.827400  
##                NA's      :15                NA's      :15                NA's      :15  
##      pvalue      padj  
## Min.      :0.0000      Min.      :0.000  
## 1st Qu.:0.4940      1st Qu.:0.976  
## Median :0.8887      Median :1.000  
## Mean   :0.7237      Mean   :0.938  
## 3rd Qu.:0.9862      3rd Qu.:1.000  
## Max.   :1.0000      Max.   :1.000  
## NA's    :15          NA's    :23470
```

```
res_sex <- res_sex %>%  
  mutate(gene = rownames(res_sex),  
         significant = ifelse((log2FoldChange > 1 & padj < 0.1), "Male",  
                             ifelse((log2FoldChange < -1 & padj < 0.1), "Female", "not significant")))  
  
universo <- res_sex$gene  
genes <- list("Female" = res_sex$gene[res_sex$significant == "Female"],  
             "Male" = res_sex$gene[res_sex$significant == "Male"])  
  
enrich_sex <- compareCluster(geneClusters = genes,
```

```

    fun = "enricher",
    TERM2GENE = term2gene,
    universe = universo,
    pAdjustMethod = "BH",
    pvalueCutoff = 0.1,
    minGSSize = 3)

ggplot(res_sex, aes(x = log2FoldChange, y = -log10(padj),
                    color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("Female" = "red",
                                "Male" = "blue",
                                "not significant" = "lightgray")) +

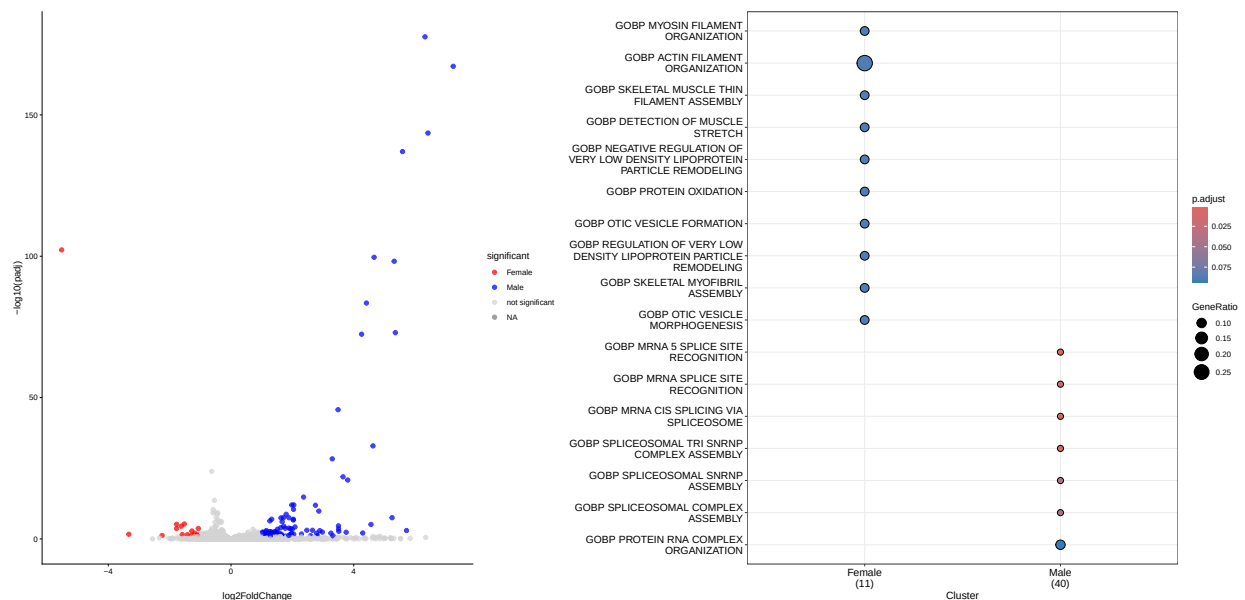
  theme_classic() +
  dotplot(enrich_sex, showCategory = 10,
          color = "p.adjust")

```

```

## Warning: Removed 23470 rows containing missing values or values outside the scale range
## ('geom_point()').

```



BRCA

Paso 1: Cargar los datos

```

brca_rna <- read.table(file="../input/brca_subset_chr1_22.tsv", sep="\t", header=TRUE, check.names = FALSE)
metadata_brca <- read.table(file="../input/brca_metadata.tsv", sep="\t", header = TRUE) %>% column_to_r

```

Paso 2: Preparar la matriz para PCA normalizando con vst y seleccionando los 5000 más variables

```
# vst
dds <- DESeqDataSetFromMatrix(
  countData = round(brca_rna),
  colData = metadata_brca,
  design = ~ 1)

## converting counts to integer mode

vsd <- vst(dds, blind = TRUE)
brca_rna_vst <- assay(vsd)

# var
gene_var <- apply(brca_rna_vst, 1, var)
top5000 <- names(sort(gene_var, decreasing = TRUE))[1:5000]
brca_rna_vst_5000 <- brca_rna_vst[top5000, ]
```

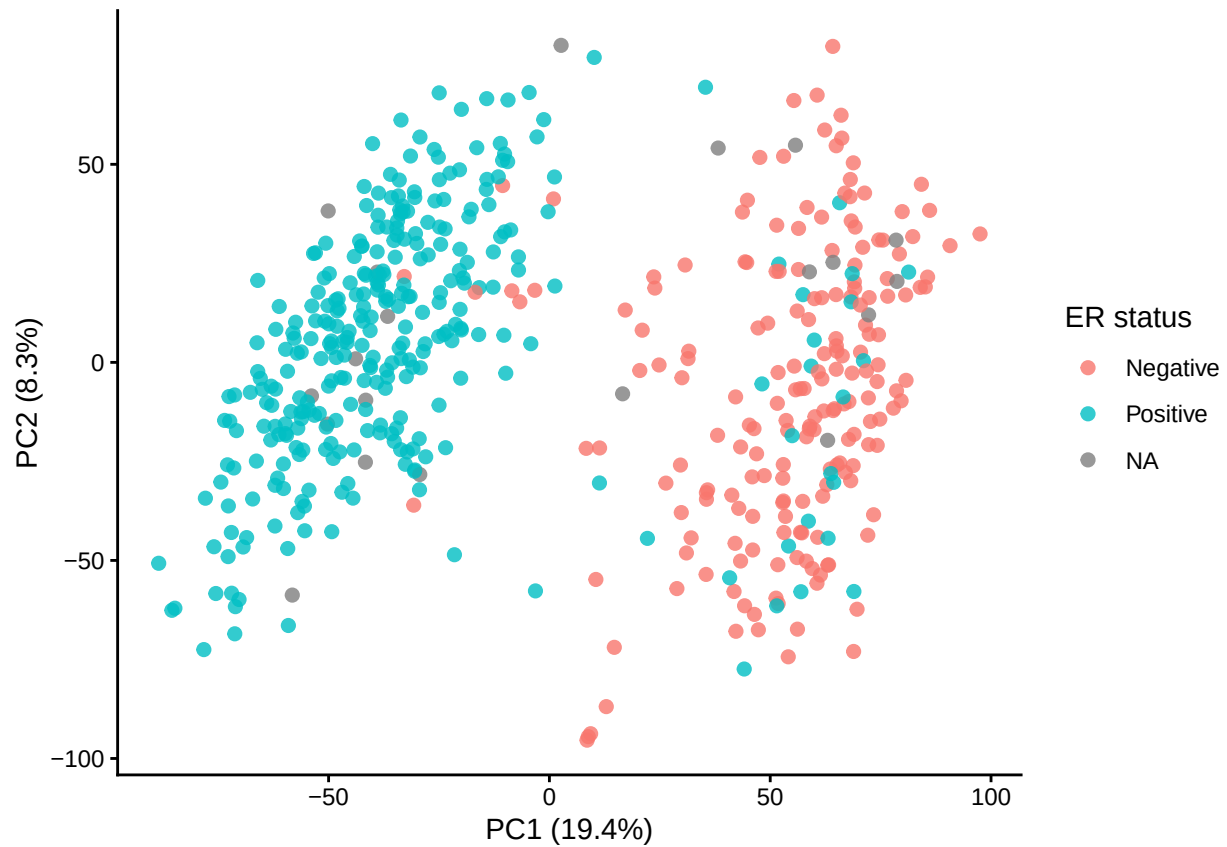
Paso 3: PCA

```
# PCA
pca_brca <- prcomp(t(brca_rna_vst_5000))
# varianza explicada
var_exp <- (pca_brca$sdev^2 / sum(pca_brca$sdev^2)) * 100
var_exp[1:10]

## [1] 19.418009  8.328388  5.395817  3.791261  3.032730  2.379081  2.047259
## [8]  1.767328  1.602311  1.409226

# pc1 vs pc2
pca_df <- as.data.frame(pca_brca$x)
pca_df$er_status <- metadata_brca[rownames(pca_df), "er_status"]

ggplot(pca_df, aes(x = PC1, y = PC2, color = er_status)) +
  geom_point(size = 2, alpha = 0.8) +
  theme_classic() +
  labs(x = paste0("PC1 (", round(var_exp[1], 1), "%)"),
       y = paste0("PC2 (", round(var_exp[2], 1), "%)"),
       color = "ER status")
```

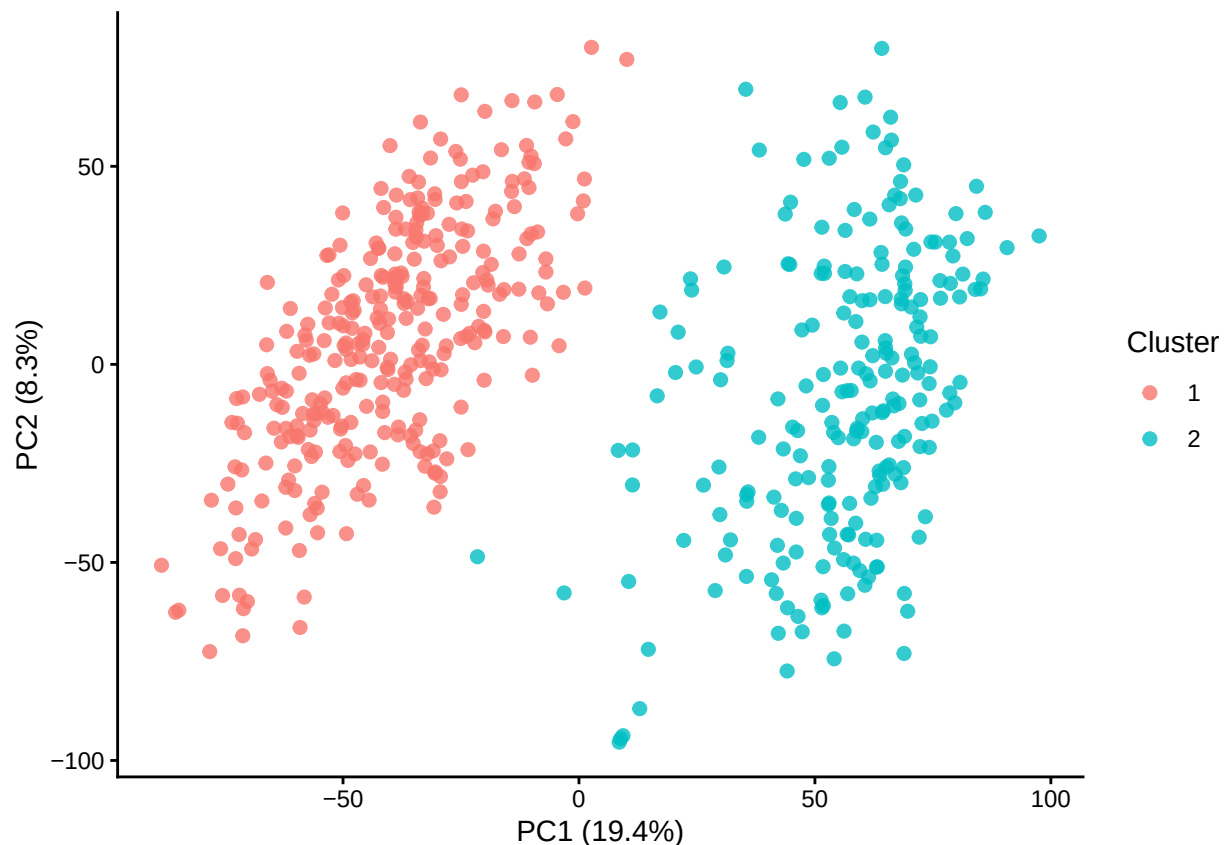


Paso 4: Clustering jerárquico

```
hc <- hclust(dist(pca_brca$x[, c("PC1", "PC2")]),
             method = "ward.D2")
cluster <- cutree(hc, k = 2)

metadata_brca$cluster <- factor(cluster[rownames(metadata_brca)])

pca_df$cluster <- metadata_brca[rownames(pca_df), "cluster"]
ggplot(pca_df, aes(PC1, PC2, color = cluster)) +
  geom_point(size = 2, alpha = 0.8) +
  theme_classic() +
  labs(x = paste0("PC1 (", round(var_exp[1], 1), "%)"),
       y = paste0("PC2 (", round(var_exp[2], 1), "%)"),
       color = "Cluster")
```



```
fisher_vars <- c("sex", "vital_status", "is_ffpe", "pr_status", "er_status")
wilcox_vars <- c("age_at_diagnosis", "days_to_death")
```

```
fisher_res <- sapply(fisher_vars, \(v)
  fisher.test(table(metadata_brca$cluster, metadata_brca[[v]]))$p.value)
```

```
wilcox_res <- sapply(wilcox_vars, \(v)
  wilcox.test(metadata_brca[[v]] ~ metadata_brca$cluster)$p.value)
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
```

```
results <- data.frame(
  variable = c(fisher_vars, wilcox_vars),
  p_value = c(fisher_res, wilcox_res) %>%
  mutate(FDR = p.adjust(p_value, method = "BH"),
    significant = FDR < 0.05)
```

```
results
```

	variable	p_value	FDR	significant
##	sex	1.426748e-01	1.664539e-01	FALSE
##	vital_status	6.135886e-01	6.135886e-01	FALSE
##	is_ffpe	2.656529e-03	5.738159e-03	TRUE
##	pr_status	3.956694e-75	1.384843e-74	TRUE
##	er_status	6.110786e-95	4.277550e-94	TRUE

```
## age_at_diagnosis age_at_diagnosis 3.278948e-03 5.738159e-03 TRUE
## days_to_death days_to_death 1.376713e-02 1.927398e-02 TRUE
```

Paso 5: Análisis de expresión diferencial

```
coldata <- metadata_brca %>% select(cluster)
brca_rna_deseq <- brca_rna[, rownames(coldata)] # sin vst

dds_brca <- DESeqDataSetFromMatrix(countData = brca_rna_deseq,
                                   colData = coldata,
                                   design = ~cluster)

dds_brca <- DESeq(dds_brca)

## estimating size factors

## estimating dispersions

## gene-wise dispersion estimates

## mean-dispersion relationship

## final dispersion estimates

## fitting model and testing

## -- replacing outliers and refitting for 9216 genes
## -- DESeq argument 'minReplicatesForReplace' = 7
## -- original counts are preserved in counts(dds)

## estimating dispersions

## fitting model and testing

resultsNames(dds_brca)

## [1] "Intercept"      "cluster_2_vs_1"

res_brca <- results(dds_brca, name = "cluster_2_vs_1") %>% as.data.frame()
summary(res_brca)

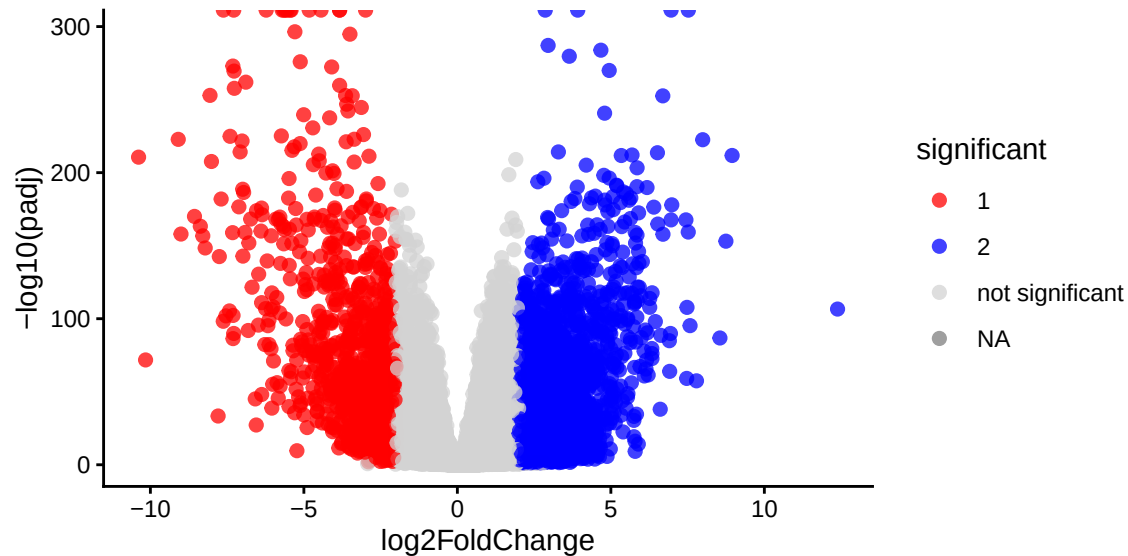
##      baseMean      log2FoldChange      lfcSE      stat
## Min.       : 0.0      Min.       :-10.38489   Min.       :0.0000   Min.       : -46.6251
## 1st Qu.: 0.4      1st Qu.: -0.04574   1st Qu.:0.0847     1st Qu.: -0.3019
## Median : 6.6      Median : 0.23875     Median :0.1667     Median : 0.8330
## Mean  : 1087.2     Mean  : 0.39387      Mean  :0.4904      Mean  : 1.6377
## 3rd Qu.: 404.2     3rd Qu.: 0.76810     3rd Qu.:0.4720     3rd Qu.: 5.1376
## Max.   :383288.4   Max.   : 12.38475     Max.   :2.9283      Max.   : 39.4700
```

```
##           NA's      :233           NA's      :233           NA's      :233
##      pvalue      padj
##  Min.   :0.00000   Min.   :0.000
## 1st Qu.:0.00000   1st Qu.:0.000
## Median :0.00117   Median :0.000
## Mean   :0.24547   Mean   :0.138
## 3rd Qu.:0.50668   3rd Qu.:0.126
## Max.   :1.00000   Max.   :1.000
## NA's    :233      NA's    :8766
```

```
res_brca <- res_brca %>%
  mutate(gene = rownames(res_brca),
         significant = ifelse((log2FoldChange > 2 & padj < 0.05), "2",
                              ifelse((log2FoldChange < -2 & padj < 0.05), "1", "not significant"))))

ggplot(res_brca, aes(x = log2FoldChange, y = -log10(padj),
                    color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("1" = "red",
                                "2" = "blue",
                                "not significant" = "lightgray")) +
  theme_classic()
```

```
## Warning: Removed 8766 rows containing missing values or values outside the scale range
## ('geom_point()').
```



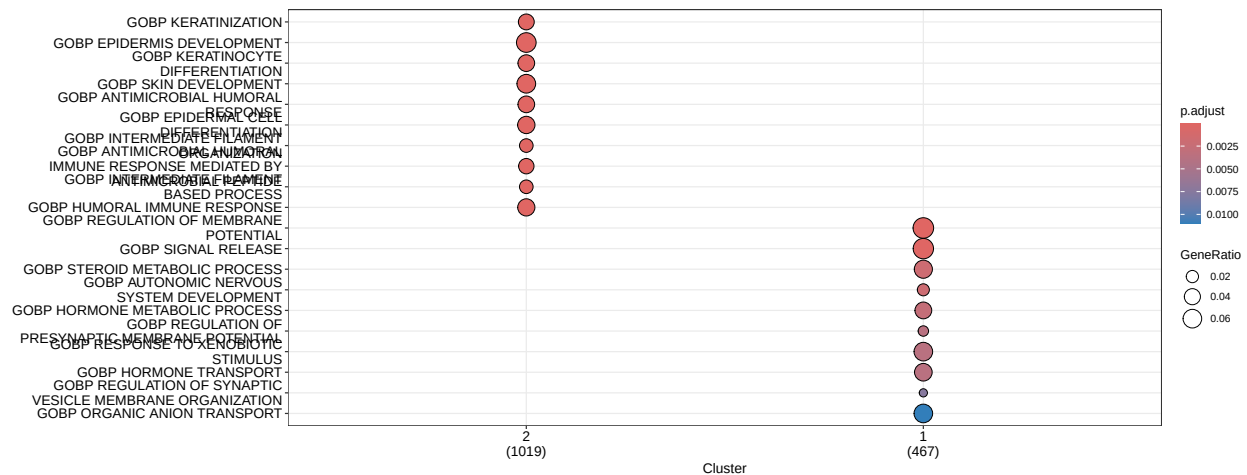
Paso 6: Enriquecimiento funcional

```
universo <- res_brca$gene
universo <- sub("\\.\\.*$", "", res_brca$gene)
genes <- list("2" = res_brca$gene[res_brca$significant == "2"],
             "1" = res_brca$gene[res_brca$significant == "1"])
```

```
genes <- lapply(genes, function(x) sub("\\..*$", "", x))

enrich_brca <- compareCluster(geneClusters = genes,
                             fun = "enricher",
                             TERM2GENE = term2gene,
                             universe = universo,
                             pAdjustMethod = "BH",
                             pvalueCutoff = 0.05,
                             minGSSize = 3)

dotplot(enrich_brca, showCategory = 10,
        color = "p.adjust")
```



El cluster 1 se enriquece en pacientes con ER y PR positivos, que a su vez se enriquece en procesos hormonales, señalización y transporte de membrana; observable en tipos luminales caracterizados por la presencia de estos receptores. Por otro lado, el cluster 2 se relaciona con programas de diferenciación epitelial como keratinización, desarrollo de epidermis, y procesos de respuesta inmune; considerando la relación con receptores hormonales del cluster 1 este coincide con un perfil no luminal.

Paso extra: enriquecimiento de los componentes principales, ¿qué captura cada componente?

```
loadings <- as.data.frame(pca_brca$rotation[, c("PC1", "PC2")])
loadings$gene <- sub("\\..*$", "", rownames(loadings))

n_top <- ceiling(nrow(loadings) * 0.05) # 5% con mayor y menor peso

genes_pc <- list(
  "PC1+" = loadings$gene[order(loadings$PC1, decreasing = TRUE)][1:n_top],
  "PC1-" = loadings$gene[order(loadings$PC1)][1:n_top],
  "PC2+" = loadings$gene[order(loadings$PC2, decreasing = TRUE)][1:n_top],
  "PC2-" = loadings$gene[order(loadings$PC2)][1:n_top])

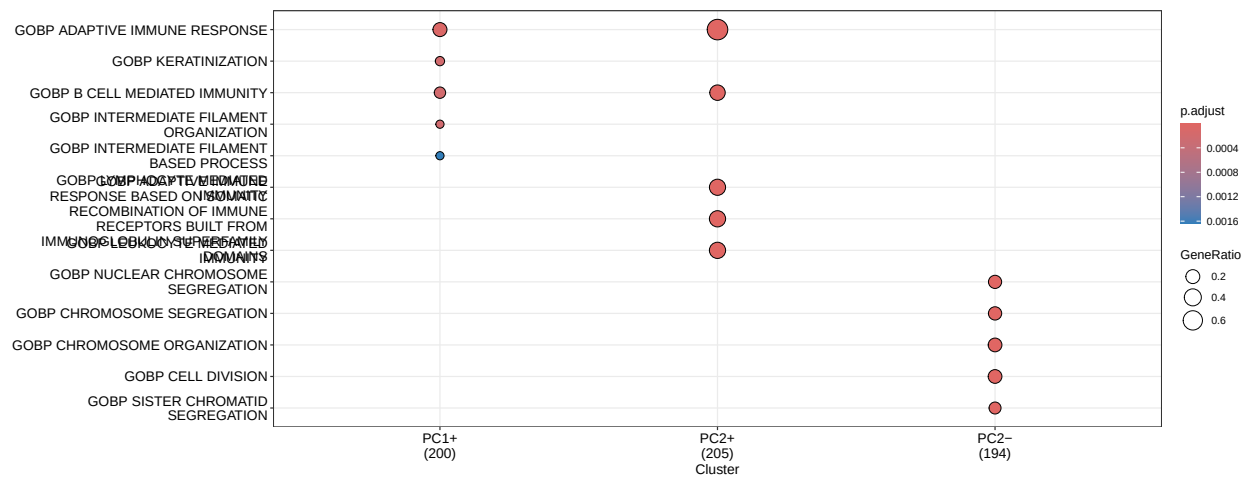
universo <- unique(loadings$gene)
enrich_pc <- compareCluster(
```

```

geneClusters = genes_pc,
fun = "enricher",
TERM2GENE = term2gene,
universe = universo,
pAdjustMethod = "BH",
pvalueCutoff = 0.05,
minGSSize = 3
)

dotplot(enrich_pc,
        showCategory = 5,
        color = "p.adjust")

```



El componente 1 captura principalmente procesos relacionados con respuesta inmune y remodelación de la matriz. Esto significa que las muestras con mayor PC1 (cluster 2) presentan perfiles de expresión mayores de estos procesos. A su vez, el componente 2 positivo se relaciona con respuesta inmune y, negativamente, con procesos de división celular. Esto podría generar un análisis de subtipos dentro de los cluster. Por lo tanto, los componentes logran capturar diversos patrones que separan a las muestras en los cluster descritos.